

Source-Locked Occurrence-to-Return Integration: Typed H1 Completion, Almost-Endpoint Möbius Progress, and a Nonduplicating Endpoint Ledger

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Abstract

We integrate the frozen TPC-153–160 structural and arithmetic outputs against the TPC-151/152 route anchors. On the structural side, the canonical cut-occurrence shadow is an L1 result, but the current-schema-only recovery route has only a scoped stop certificate. The selected augmented route is not testable because a theorem-backed production occurrence-provenance crosswalk is absent. We encode H1 as an explicit map/scalar disjunction. The map clause contains nine separate defects and an independent occurrence registry; the scalar clause contains both a complete frontier estimate and a growing-scale eligible-tail disposition.

On the arithmetic side, the strongest imported advance is the TPC-159 actual periodic-core prefix theorem outside a sparse dyadic shadow, classified as L1 actual-prefix almost-endpoint. TPC-157, TPC-158, and TPC-160 give positive weight, phase, and Abel-return interfaces, but the production weight, phase, normalization, and atomic endpoint registries remain unavailable. A typed proof DAG records evidence, scope, carrier, normalization, and artifact readiness, computes clause-wise minimal missing antichains and parent-ready open frontiers, and enforces a transitive arithmetic firewall at H9. A third-generation endpoint ledger gives each of six listed charges a unique owner; it does not declare the unknown physical registry complete. The resulting verdict is `NOT_TESTABLE`: there is no positive fixed- X -power L2 result, 1/400 endpoint, prime-pair lower bound, or twin-prime theorem.

1 Integration statement

The earlier integration snapshot located the first unsupported object at a frontier occurrence lift [11, 8]. The next structural papers sharpened that pointer: the archive determines a canonical shadow, but not theorem-backed downstream occurrence provenance [2, 3, 12, 7]. In parallel, the determinant-two Möbius corridor acquired weight, phase, prefix, and Abel interfaces [9, 5, 4, 6]. The purpose of the present paper is to assemble these two lines without allowing an interface theorem to fill a missing physical artifact.

Theorem 1.1 (Source-locked occurrence-to-return integration). *For the frozen TPC-151–160 source snapshot, the following statements hold.*

- (i) *The current-schema-only canonical-lift route is stopped in its declared formal-schema cell, while the theorem-augmented map route is not stopped.*
- (ii) *On the selected map route the unique minimal NT-ancestor is*

H1.theorem_backed_occurrence_provenance_crosswalk.

On the scalar route the minimal antichain consists of the complete FUM $o(X)$ theorem and the theorem-backed ETO disposition.

- (iii) *The H9 physical-registry node has no direct or transitive arithmetic ancestor.*
- (iv) *The strongest arithmetic import is the TPC-159 theorem, classified as L1 actual-prefix almost-endpoint. The production weight, phase, endpoint, and normalization registries are all NT.*
- (v) *The arithmetic fixed-X exponent is zero and the physical loss is unknown. Consequently the 1/400 endpoint is unpaid.*

Thus the selected synthesis verdict is `NOT_TESTABLE`, not `GO`, `STOP`, or architecture infeasibility.

The theorem is an integration statement: its mathematical inputs are the cited source results, and its machine component verifies their exact declared fields and the consistency of the composed contract. A source hash is an integrity lock, never theorem evidence.

2 Frozen structural inputs

2.1 The shadow and the scoped obstruction

Let $\mathcal{C}_X^{\text{ns}}$ denote the archived nonsoft cut paths. It is the declared disjoint union

$$\mathcal{C}_X^{\text{ns}} = \mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}},$$

where ETO denotes *eligible-tail open* and FUM denotes *frontier unmapped*. TPC-153 constructs one sourced unit shadow row for each archived nonsoft path. The resulting identity injection is row-separated and conservative, but it contains no actual occurrence identifier, branch count, canonical parent, later stage, physical group, or active-support certificate [2].

The finite production census is

$$\#\mathcal{C}_X^{\text{ETO}} = 0, \quad \#\mathcal{C}_X^{\text{FUM}} = 2988.$$

The first equality is a finite archive fact, not an asymptotic ETO disposition theorem.

TPC-154 exhibits two distinct *formal* conservative completions of every shadow row—one child of weight 1, or two children of weight 1/2—that forget to the same shadow [3]. This proves nonuniqueness in the maximal current schema class. It does not assert that both completions occur on the actual arithmetic carrier. Hence

$$\text{current-schema-only lift} = \text{STOP}_{\text{scoped}}, \quad \text{augmented actual lift} = \text{NT}.$$

TPC-155 supplies a sound finite verifier for a proposed occurrence bundle, including quotient, grouping, selector, cover, reconnection, and registry exports [12]. Its positive fixture is synthetic L0 evidence. No production bundle has been supplied.

2.2 The exact H1 disjunction

Write C_{prov} for the theorem-backed cut-to-occurrence provenance crosswalk and let

$$\mathbf{D} = (D_L, D_{QD}, D_{QZ}, D_G, D_P, D_{DZ}, D_{GP}, D_{\text{cover}}, D_{\text{rec}}).$$

The occurrence registry R_{occ} is deliberately not renamed D_{occ} or absorbed into $\mathbf{D}$. It is a separate positive totality object.

Definition 2.1 (Typed H1 contract). The two sufficient clauses are

$$\mathcal{M}_{H1} = C_{\text{prov}} \wedge \bigwedge_{\Delta \in \mathcal{D}} (\Delta = 0) \wedge R_{\text{occ}}, \quad (1)$$

$$\mathcal{S}_{H1} = [S_{\text{FUM}}(X) = o(X) \text{ on the complete original carrier}] \wedge E_{\text{ETO}}, \quad (2)$$

$$H1 = \mathcal{M}_{H1} \vee \mathcal{S}_{H1}. \quad (3)$$

Here E_{ETO} is a growing-scale theorem-backed totalization or complete softness statement for every ETO path.

This formulation prevents three compressions: a shadow is not an actual lift; nine map defects are not a single unnamed “four-map check”; and a frontier scalar estimate does not dispose of ETO.

Proposition 2.2 (Clause-wise minimal missing antichains). *In the frozen snapshot,*

$$\begin{aligned} \min \text{NT}(\mathcal{M}_{H1}) &= \{C_{\text{prov}}\}, \\ \min \text{NT}(\mathcal{S}_{H1}) &= \{S_{\text{FUM}}(X) = o(X), E_{\text{ETO}}\}. \end{aligned}$$

All nine defect evaluations and the occurrence registry are blocked downstream of C_{prov} ; they are not promoted to zero.

Proof. The crosswalk is the first unavailable artifact on every selected map branch. The lift, quotient, grouping, selector, cover, reconnection, and registry nodes therefore have readiness `BLOCKED_BY_PARENT`. Removing descendants of an earlier unavailable ancestor leaves the singleton map antichain. The two scalar conjuncts are independent missing leaves, so both remain. $\square$

3 Typed DAG and readiness semantics

Each proof node is stored as the tuple

$$(\text{id}, \text{type}, \text{status}, \text{role}, \text{level}, \text{parents}, \text{clauses}, \\ \text{evidence}, \text{scope}, \text{carrier}, \text{normalization}, \text{artifact}, \text{readiness}).$$

Allowed dependency types are evidence, target, all, any-clause, and scoped-stop. Allowed readiness states distinguish `READY`, `MISSING`, `BLOCKED_BY_PARENT`, and `NOT_APPLICABLE`. Thus the statement “not proved” is not silently identified with “testable on the current data.”

The manifest contains 53 typed nodes and 12 exact imports. Every import and node references a registered scope, carrier, normalization, and artifact. For an all-node the status is derived from every parent; for an any-clause node it is derived clause by clause. The graph is checked for cycles before any verdict is computed.

Definition 3.1 (Parent-ready open frontier). An open target is parent-ready if its required artifact is ready and all of its declared parents are proved.

Exactly two targets satisfy this definition:

1. a direct additive-twist theorem on the existing actual determinant-two core; and
2. a pointwise theorem at endpoints inside the dyadic bad set.

They are genuine arithmetic research frontiers, but neither supplies a production occurrence crosswalk or physical registry.

3.1 The H9 transitive firewall

Let $\text{Anc}(v)$ be all direct and transitive ancestors of a DAG node. The integration enforces

$$\text{Anc}(\text{H9}) \cap \{\text{arithmetic-core, arithmetic-target, arithmetic-negative roles}\} = \emptyset. \quad (4)$$

H9 may depend on structural occurrence, cover, and reconnection records and on literal physical registries. It may not inherit TPC-157–160 as evidence for those registries. The frozen H9 ancestor set passes (4). This is stronger than checking only the immediate parents.

4 The actual-core arithmetic corridor

We now record exactly what the arithmetic line contributes. Let

$$q = as, \quad t(z) = ad + qz, \quad c(z) = \mu(d + sz)\mu(u + az),$$

with $su - ad = 2$ and the small-polylogarithmic modulus conditions from TPC-149. The quantitative affine-correlation input ultimately comes from Tao–Teräväinen [1]; the actual periodic Möbius corridor was isolated in TPC-149 [10]. Throughout this section,

$$\mathcal{L} := \log X, \quad e(x) := \exp(2\pi i x).$$

4.1 Weight and phase interfaces

For $I_N = \{z : N < t(z) \leq 2N\}$, TPC-157 proves outside the same exceptional set that

$$\frac{q}{N} \left| \sum_{z \in I_N} c(z)w(z) \right| \ll \inf_{\substack{\rho \text{ period } R \\ qR \leq \mathcal{L}^{\eta_0}}} \left[\|\rho\|_\infty \mathcal{L}^{-\kappa_0} + \frac{q}{N} \sum_{z \in I_N} |w(z) - \rho(z)| \right]. \quad (5)$$

This is a positive L1 actual-core weight *interface*. A literal physical theorem additionally needs a source-locked production w and a decaying approximation cost.

TPC-158 applies (5) on a major-arc cell. If

$$qR \leq \mathcal{L}^{\eta_0}, \quad L \left| \alpha - \frac{k}{R} \right| \leq \mathcal{L}^{-A},$$

then

$$\frac{q}{N} \left| \sum_{z \in I_N} c(z)e(-\alpha z) \right| \ll \mathcal{L}^{-\kappa_0} + \mathcal{L}^{-A}. \quad (6)$$

For uniformly separated minor-arc cells, TPC-158 proves only that the small-period approximation route fails. It neither proves that the Möbius correlation is large nor stops a direct additive-twist theorem.

4.2 Almost-endpoint prefixes

Let $\mathcal{E}_X^*$ be the source exceptional-scale set and put

$$J = \lceil A \log_2 \log X \rceil, \quad \mathcal{S}_{X,J} = \bigcup_{1 \leq j \leq J} 2^j \mathcal{E}_X^*.$$

TPC-159 performs exact dyadic telescoping. For

$$2^J \sqrt{X} \leq T \leq X, \quad T \notin \mathcal{S}_{X,J},$$

it obtains

$$\frac{q}{T} \left| \sum_{0 < t(\tilde{z}) \leq T} c(z) \rho(z) \right| \ll \|\rho\|_\infty \left(\mathcal{L}^{-\kappa_0} + 2^{-J} + \frac{q}{T} \right). \quad (7)$$

There is no factor J in the correlation term because the dyadic block masses form a geometric sum. Meanwhile

$$\frac{1}{\log X} \int_{\mathcal{S}_{X,J}} \frac{dT}{T} \ll J \mathcal{L}^{-\kappa_0} = \mathcal{L}^{-\kappa_0 + o(1)}.$$

This moves the actual periodic core from source-native dyadic blocks to cumulative prefixes outside a sparse shadow. It is therefore the strongest arithmetic export in the present integration.

It is not an all-endpoint statement. A predetermined atomic endpoint can lie in a continuously sparse set.

4.3 Abel return and its exact boundary

For ordered points $t_1 < \dots < t_m \leq T$, put

$$\sigma_i = c(z_i) \rho(z_i), \quad A_i = \sum_{j \leq i} \sigma_j, \quad d_i = \begin{cases} w_i - w_{i+1}, & i < m, \\ w_m, & i = m. \end{cases}$$

Then the exact identity

$$\sum_{i=1}^m \sigma_i w_i = \sum_{i=1}^m A_i d_i$$

and (7) yield the TPC-160 interface

$$\frac{q}{T} \left| \sum_i \sigma_i w_i \right| \ll \|\rho\|_\infty \left[\varepsilon_X V_{\text{good}}(w) + \left(1 + \frac{q}{T} \right) V_{\text{bad}}(w) \right], \quad (8)$$

where

$$\varepsilon_X = \mathcal{L}^{-\kappa_0} + 2^{-J} + \frac{q}{T}.$$

For a prefix cutoff the Abel derivative is one atom at the selected endpoint. Continuous exceptional density cannot certify that this atom avoids the bad set. Hence TPC-160 is a positive weighted almost-endpoint interface, but not a production physical return.

5 Endpoint Ledger V3

Earlier ledgers separated arithmetic and physical losses. Version V3 additionally assigns a unique owner to each of six listed charges. The full-synthesis ledger references its two child ledgers and carries no listed charge a second time. This is a nonduplication contract, not a claim that the unknown physical-loss registry is complete.

Charge	Owner	Quantity	State
correlation log saving	arithmetic	log-amplitude saving	proved
dyadic shadow	arithmetic	log exceptional measure	proved
good variation	physical	log-amplitude cost	not testable
bad variation	physical	atomic endpoint cost	not testable
phase return	physical	phase-return cost	not testable
four-sign reconnection	physical	reconnection cost	not testable

Table 1: Unique ownership for the six listed endpoint charges.

The arithmetic inputs provide powers of $\log X$, so their certified fixed- X exponent is

$$\sigma_{\text{arith}}^{(X)} = 0.$$

The physical upper and lower exponents remain unknown and unknown is not charged as zero. At amplitude scale the target remains

$$\sigma_{\text{raw}}^{(X)} \geq \frac{1}{400}, \quad \Lambda_{\text{phys}}^{(X)} < \frac{1}{400}, \quad \sigma_{\text{raw}}^{(X)} - \Lambda_{\text{phys}}^{(X)} > 0.$$

No power of $\log X$ pays this fixed- X budget. Consequently the arithmetic, physical, and full-synthesis ledgers all remain incomplete.

6 Route decision and next exact objects

The route table now has five cells.

1. *Current-schema-only lift*: scoped stop.
2. *Occurrence-augmented map*: selected, NT, first missing C_{prov} .
3. *Scalar plus ETO*: open but NT, with two independent missing leaves.
4. *Direct additive twist on the actual core*: parent-ready OPEN.
5. *Pointwise bad-endpoint theorem on the actual core*: parent-ready OPEN.

The first two arithmetic open targets can improve the core even if the structural crosswalk remains unavailable. Conversely, a future crosswalk can complete H1 without solving the additive-twist problem. This separation is intentional: it allows progress on either line without relabeling one as progress on the other.

For the selected full synthesis, the minimal unavailable artifact set is

$$\left\{ \begin{array}{l} C_{\text{prov}}, \text{ literal weight registry, phase-cell registry,} \\ \text{atomic endpoint registry, physical normalization registry} \end{array} \right\}.$$

The two parent-ready arithmetic targets are recorded separately as open dependencies, not as missing production artifacts.

7 Machine certificate and claim boundary

The standard-library audit uses `CANONICAL_UTF8_LF_V2` to lock 24 scripts and outputs. It validates:

- exact frozen source statuses and claim boundaries;
- strict JSON schemas with no undeclared object fields;
- the acyclic 53-node ALL/ANY-clause DAG;
- the nine defects and independent occurrence registry;
- clause-wise minimal NT antichains;
- the two parent-ready OPEN frontiers;
- the transitive H9 arithmetic firewall; and
- unique ownership of the six listed charges in Endpoint Ledger V3.

Mutation regressions reject shadow-to-lift promotion, defect omission, registry compression, an arithmetic H9 ancestor, duplicate or omitted members of the six-charge contract, log-to- X -power conversion, and unsupported L2 or 1/400 claims.

The status **PASS** means that this integration certificate is internally consistent with its source-locked inputs. It does not mean that a production occurrence lift exists, that a generic phase is controlled, that every endpoint is good, or that the full physical H3 packet has a saving. In particular, this paper proves no positive fixed- X -power L2 estimate, no $B_{h_0, \delta}(X) = o(X)$ theorem, no prime-pair lower bound, and no twin-prime theorem.

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